Optimizing Conversions: Swiggy Web Traffic Analysis

GitHub	https://bit.ly/40IXYJf
Tableau	https://bit.ly/4biLbTW

🔗 Live Dashboard

👉 https://bit.ly/4biLbTW

Swiggy Web Traffic & Conversion Performance Analysis - Analytical Report

1. Executive Summary

Swiggy experiences high levels of website traffic across multiple acquisition channels and devices. Despite strong traffic volumes, conversion performance required deeper evaluation to determine whether user drop-offs were driven by traffic quality, engagement issues, or experience-related factors.

This analysis evaluates **session-level web traffic data** to assess conversion efficiency, user engagement, and behavioral patterns across channels, devices, and user segments. An interactive Tableau dashboard was developed to enable stakeholders to monitor key performance indicators (KPIs) and perform targeted drill-downs for data-driven decision-making.

The analysis indicates that while conversion rates remain stable across segments, **bounce rates show greater variability**, suggesting opportunities to improve traffic targeting and early-stage user experience rather than checkout or conversion mechanics.

2. Business Context & Problem Statement

Swiggy's digital platform serves a large and diverse user base. Marketing and product teams require visibility into:

- How different traffic sources contribute to meaningful engagement
- Whether device type impacts user behavior and conversion outcomes
- Which user segments exhibit higher drop-offs during early session stages
- How performance metrics evolve over time

The absence of a consolidated performance view made it difficult to distinguish between **high-volume traffic** and **high-quality traffic**, leading to potential inefficiencies in marketing spend and optimization efforts.

3. Objectives of the Analysis

The primary objectives of this analysis were to:

1. Measure overall traffic and conversion performance using session-level KPIs
2. Evaluate traffic quality through bounce rate analysis
3. Compare conversion efficiency across traffic sources and device types
4. Identify behavioral differences between new and returning users
5. Track performance trends over time to detect anomalies or seasonal impacts
6. Provide an interactive dashboard enabling flexible stakeholder exploration

4. Data Scope & Key Metrics

Data Scope

The dataset consists of aggregated website session data including:

- Traffic source
- Device type
- User type (New vs Returning)
- Session engagement indicators
- Conversion outcomes
- Daily activity trends

Key Performance Indicators

The following KPIs were selected to balance **volume, quality, and efficiency**:

- **Total Website Sessions (Last 90 Days)**
- **Session Conversion Rate**
- **Session Bounce Rate**

- **Average Time on Site**
- **Daily Sessions Trend**
- **Daily Conversion Rate Trend**

These metrics were chosen to ensure clarity, relevance, and alignment with business decision-making.

---

5. Analytical Approach

The analysis followed a structured approach:

- Developed an executive-level dashboard using Tableau Public with a clear analytical flow
- Segmented performance by **traffic source**, **device type**, and **user type**
- Avoided misleading dual-axis charts by separating volume metrics from efficiency metrics
- Enabled interactive filters to support targeted investigation without recreating views
- Designed visuals with a focus on readability, consistency, and stakeholder usability

Special attention was given to maintaining **metric integrity**, ensuring session-level calculations were not misrepresented through aggregation or visualization choices.

---

6. Key Findings & Insights

6.1 Traffic & Conversion Performance

- Session conversion rates remain largely consistent across traffic sources, averaging approximately **40–41%**
- No significant conversion disparity is observed between desktop, mobile, and tablet users

6.2 Traffic Quality & Engagement

- Bounce rates exhibit greater variation than conversion rates across channels
- Certain acquisition sources and device types show elevated bounce rates, indicating lower-quality traffic or early-stage engagement issues
- Mobile users demonstrate slightly higher bounce rates, though conversion rates remain comparable to desktop users

6.3 Trend Analysis

- Daily sessions and conversion rates show overall stability across the analysis period

- Occasional dips suggest possible campaign-driven or seasonal influences rather than systemic performance issues

7. Conclusions

The analysis indicates that Swiggy's conversion performance is **structurally sound** across devices and channels. However, variability in bounce rates highlights opportunities to improve:

- Traffic targeting strategies
- Landing page relevance
- Early user experience and onboarding flows

The data suggests that improving **traffic quality and engagement** is likely to yield higher returns than increasing acquisition volume alone.

8. Recommendations

Based on the analysis, the following actions are recommended:

1. Optimize acquisition strategies to reduce low-quality traffic contributing to higher bounce rates
2. Conduct UX and landing page evaluations for segments with elevated early exits
3. Prioritize engagement improvements for mobile users to reduce initial drop-offs
4. Continue monitoring trends through an interactive dashboard to assess the impact of optimizations

9. Tools & Techniques Used

- Tableau Public for dashboard development and visualization
- Web analytics and session-based KPI design
- Segmentation and comparative analysis
- Data storytelling and stakeholder-focused reporting

10. Business Impact

This project demonstrates how structured analytics and dashboard storytelling can transform raw web traffic data into **actionable insights**. The resulting dashboard equips stakeholders with the ability to:

- Distinguish between traffic volume and traffic quality
- Identify optimization opportunities with minimal analytical overhead
- Support data-driven marketing and product decisions